

Git Quick Reference — Clone, Branches & Commands

1. Clone a Repository

Basic clone (most common)

```
pwd                # see where you are
cd /c/Users/kenneth/Projects # go where you want the repo
git clone https://github.com/username/repo-name.git
cd repo-name
ls
```

Creates a folder repo-name with all files and history. **No git init needed.**

Clone into a folder you already made

```
mkdir my-project
cd my-project
git clone https://github.com/username/repo-name.git .
```

The . means "clone into this folder." Folder must be **empty**. Still **no git init**.

Clone a specific branch directly

```
git clone -b hello https://github.com/username/repo-name.git
```

Lands you on branch hello immediately. Branch names are **case-sensitive** (hello ≠ Hello).

2. Know Where You Are

Command	What it does
pwd	Show current folder
ls	List files
ls -la	List all files including hidden (.git)
cd folder	Go into a folder
cd ..	Go up one level
cd ~	Go to home folder

Important: Git commands only work **inside the repo folder** (where `.git` exists).

```
Wrong:  ~/skyscraper           (parent folder, not a repo)
Right:  ~/skyscraper/Skycrapers (actual repo)
```

Error if you're in the wrong place: fatal: not a git repository

3. Branches

See branches

```
git branch          # local branches (* = current)
git branch -a       # local + remote branches
git status          # shows current branch at top
```

Switch to an existing branch (already on GitHub)

```
git checkout hello
# or
git switch hello
```

Create a NEW branch and switch to it

```
git checkout -b my-new-branch
# or
git switch -c my-new-branch
```

Push a branch to GitHub

```
git push -u origin hello
```

4. Daily Workflow

```
cd /path/to/repo          # 1. go into repo
git status                 # 2. see branch + changes
git pull                   # 3. get latest from GitHub
# ... edit files ...
git add filename           # 4. stage specific file (safer)
git commit -m "Your message" # 5. save locally
git push                   # 6. send to GitHub
```

5. Git Command List

Setup & Clone

Command	What it does
git --version	Check Git is installed
git clone <url>	Copy repo from GitHub
git clone -b <branch> <url>	Clone and checkout specific branch
git init	Create new empty repo (don't use with clone)

Navigation & Status

Command	What it does
pwd	Show current folder
git status	Branch, modified/staged/untracked files
git log	Commit history
git log --oneline	Short commit history
git diff	See unstaged changes
git diff --staged	See staged changes

Branches

Command	What it does
git branch	List local branches
git branch -a	List all branches (local + remote)
git checkout <branch>	Switch to existing branch
git checkout -b <branch>	Create new branch and switch
git switch <branch>	Switch branch (newer syntax)
git switch -c <branch>	Create + switch (newer syntax)
git branch -d <branch>	Delete local branch

Save Changes

Command	What it does
git add <file>	Stage one file

<code>git add .</code>	Stage everything (use carefully!)
<code>git reset</code>	Unstage all files
<code>git commit -m "msg"</code>	Commit staged changes
<code>git restore <file></code>	Discard unstaged changes to file

Remote (GitHub)

Command	What it does
<code>git pull</code>	Download + merge latest from GitHub
<code>git push</code>	Upload commits to GitHub
<code>git push -u origin <branch></code>	Push new branch and set tracking
<code>git fetch</code>	Download remote info without merging
<code>git remote -v</code>	Show remote URLs

6. File Status Meanings

Status	Meaning
modified	Existing file changed
untracked	New file Git doesn't track yet
staged	Ready to commit (<code>git add</code> done)
committed	Saved in Git history

IDE labels: **M** = modified, **A** = added, **D** = deleted

7. Things to Avoid

Don't	Do instead
<code>git init</code> then <code>git clone</code>	Just <code>git clone <url></code>
<code>git add .</code> blindly	<code>git add specific-file</code>
Commit <code>.iml</code> / <code>.idea/</code> files	Ignore them (IntelliJ config)
Run <code>git</code> in parent folder	<code>cd</code> into repo with <code>.git</code>
<code>git checkout -b hello</code> if branch exists	<code>git checkout hello</code>

8. One-Page Example (Start to Finish)

```
# Clone
cd ~/Projects
git clone -b hello https://github.com/username/Skycrapers.git
cd Skycrapers

# Check state
pwd
git status
git branch

# Work on a new branch
git checkout -b my-feature
# edit files...
git add topics/some-file/README.md
git commit -m "Update README"
git push -u origin my-feature
```

Quick Reference Card

CLONE	<code>git clone <url></code> <code>git clone -b <branch> <url></code>
WHERE	<code>pwd / cd <folder> / ls</code>
BRANCH	<code>git branch</code> <code>git checkout <branch></code> (switch existing) <code>git checkout -b <branch></code> (create new)
WORKFLOW	<code>git status</code> → <code>git add</code> → <code>git commit</code> → <code>git push</code>